Ep 10 · Evals & Observability — Cheat Sheet

AI System Design · AI Engineering with Ajit · github.com/byruajit/ai-system-design

TWO LOOPS

Offline: golden set → metrics → CI gate → canary **Online:** traces → sampled review → feedback → alerts → **new golden items**

METRICS BY LAYER

Layer	Metric
Retrieval	context recall / precision @k
Answer	correctness · faithfulness
Safety	refusal on unanswerable · **ACL leak = 0** · injection pass rate
Ops	TTFT · p95 · tokens · cost/answer
Users	thumbs · escalations · re-asks

GOLDEN SET

50–100 real questions → expected sources + required facts + labels (answerable / unanswerable / restricted / adversarial). In Git. Grows weekly.

LLM-AS-JUDGE

Calibrate vs ~100 human labels · pin the judge version · deterministic checks first

CI GATE

Path-filtered trigger · deterministic metrics block merges · judged metrics nightly · traces as artifacts

TRACING (PER REQUEST)

Spans: retrieval · rerank · LLM · tools. Attributes: model + prompt version, tokens, cost, chunk IDs. **Redact payloads.**

ALERT ON QUALITY

Refusal-rate spike · faithfulness drop · cost/answer drift · retrieval score shift

OWNERSHIP

Product: pipeline + gate · Domain experts: answers · Risk/audit: safety tests + thresholds